***CONDITIONAL PROBABILITY***:

$$P(E|F) = \frac{|E \cap F|}{|F|} = \frac{|E \cap F|/|S|}{|F|/|S|}$$

Noting $P(E \cap F) = P(F \cap E)$, we have:

$$P(E \cap F) = P(F \cap E) = P(F)P(E|F) = P(E)P(F|E)$$

***TOTAL PROBABILITY:*** $\mathrm{E} = \mathrm{E} \cap \mathrm{S} = \mathrm{E} \cap (\mathrm{F} \cup F^C) =$
$= (E \cap F) \cup (E \cap F^C)$

*The probability E is given by:*

$$P(E) = P(E|F)P(F) + P(E|F^C)P(F^C)$$

***BAYES THEOREM:*** Let mutually exclusive events $F_1, F_2, \ldots\ldots, F_N$ be a partition of sample space. Then, $= \frac{P(E|F_i)P(F_i)}{\sum_{i=1}^{N} P(E|F_i)P(F_i)}$

## *INDEPENDENCE*

***Definiton:*** *Two events are E and F said to be independent if*

$$P(E \cap F) = P(E)P(F) \rightarrow P(E|F) = P(E), P(F|E) = P(F)$$

*Three events E, F, and G are said to be independent if*

$$P(E \cap F \cap G) = P(E)P(F)(G)$$
$$P(E \cap F) = P(E)P(F)$$
$$P(E \cap G) = P(E)P(G)$$
$$P(F \cap G) = P(F)P(G)$$

**Counter example:** Consider two independent tosses of a coin.

- Let A={heads on toss 1}, B={heads on toss 2}, C={two tosses are equal}
- P(A)=P(B)=1/2
- $P(C) = P(A \cap B) + P(A^C \cap B^C) =$ (1/4)+(1/4)= 1/2
- A and B are independent by the definition.
- P(C|A)=P(B|A)=P(B)= 1/2 =P(C)
- So, A, B and C are pairwise
  - $P(A \cap B \cap C) = P(A \cap B) = P(A)P(B) = 1/4$
  - P(A)P(B)P(C) = 1/8
- So, A,B,C are not independent from each other.

***RANDOM VARIABLE:***

***Definition:*** *A random variable is a real-valued function of the elements of a sample space, S.*

*If X takes on a finite or countably infinite number of values, X is called as a discrete random variable.*

***DISCRETE RANDOM VARIABLES***

***Definition:*** *The probability mass function (pmf) of a discrete random variable X is a function that assigns a probability to each possible value of X and is denoted by p(x)=P(X=x). Assume that X takes the values x1, x2,…*

$$\sum_{i=1}^{\infty} p(X_i) = 1$$

***Definition:*** *The cumulative distribution function (cdf) specifies the random variable is less than equal to x.*

$F(X) = P(X \leq x) \qquad -\infty < x < \infty$

$P(a \leq X \leq b) = F(b) - F(a), for\ all\ a < b$

*F is non-decreasing function, i.e., if a<b, then* $F(a) \leq F(b)$

*P(x=1) = 1/4*
*P(x=2) = 1/2*
*P(x=3) = 1/8*
*P(x=4) = 1/8*

$$F(X) = \begin{cases} 0, & x<1 \\ 1/4, & 1 \leq x < 2 \\ 3/4, & 2 \leq x < 3 \\ 7/8, & 3 \leq x < 4 \\ 1, & x > 4 \end{cases}$$

***EXPECTATION: (In other words -> first moment, average, mean)***

*İstatistiki dağılımına göre rastgele bir değişken verildiğinde, temel özellikleri uygun bir şekilde belirlenmiş önlemlerle özetlemek faydalı olacaktır.*

*Ayrık rasgele değişken X'in beklenen değeri, X'in alabileceği olası değerlerin ağırlıklı ortalamasıdır, her değer ilgili olasılık tarafından ağırlıklandırılır.*

$$\boldsymbol{E[X] = \sum_i x_i P(X = x_i) = \sum_i x_i p(x_i)}$$

**Ex:** Our average winning per game / Rolling a fair die

**Example:** Let $X$ denote a random variable that takes the values of -1, 0, and 1 with respective probabilities of 0.2, 0.5, and 0.3. Calculate $E[X^2]$.

Define a new random variable $Y=X^2$.

$X = -1 \rightarrow Y = 1$
$X = 0 \rightarrow Y = 0$
$X = 1 \rightarrow Y = 1$

$$P(Y=1) = P(X=-1) + P(X=1) = 0.5$$

$$P(Y=0) = P(X=0) = 0.5$$

$$E(Y) = y_1 P(Y=y_1) + x_2 P(Y=y_2)$$
$$= 1(0.5) + 0(0.5) = 0.5$$

**Proposition:** *Assume X is a discreate random variable. The expected value of g(x) where g(.) is a real-valued function is given by,*

$$E[g(X)] = \sum_i g(x_i)P(X = x_i)$$

***Definition:*** *$E(X^n)$ is called the $n^{th}$ moment of X. Using the proposition;*

$$E(X^n) = \sum_i x_i^n P(X = x_i)$$

- ✓ ***A function is said to be linear operation if***

$$\boldsymbol{f(a_1z_1 + a_2z_2 + \cdots\ a_nz_n) = a_1f(z_1) + a_2f(z_2) + \cdots + a_nf(z_n)}$$

- ✓ ***Expectation is a linear operation,***

***E[aX+b] = a E[X]+b***

***VARIANCE***

*If X is a random variable with mean E[X], the variance of X is defined by,*

***Var(X) = E[x²] – E[X]²***

*The square root of Var(X) is called the standard deviation of X and denoted by SD(X).*

***SD(X)=***$\sqrt{Var(X)}$

Linear operation variance

**Var(a X+ b) = a² Var(X)**

For independent random variables X and Y and constants a and b, we have

**Var(a X+b Y)= a² Var(X) + b² Var(X)**

❖ ***Discrete random variables are often classified according to their pmf.***

| Distribution | Parameters | Situation |
|---|---|---|
| Discrete Uniform | $(x_1,x_2,\ldots,x_n)$ | Each element of support has equal probability |
| Bernoulli | β | 1 trial resulting in Success or Failure |
| Binomial | N, β | (fixed) n Bernoulli trials, results independent, # successes unknown |
| Poisson | X | large n, small p, $np \rightarrow \lambda$, # successes unknown |
| Negative Binomial | r, β | unknown # trials, # successes known |
| Geometric | P | Neg. Binomial with k = 1. |
| Hypergeometric | N, m, n | # of objects from the group with m elements out of n randomly chosen elements |

**DISCRETE UNIFORM RANDOM VARIABLE**

Random variable X takes values $x_1, x_2, \ldots, x_n$ with equal probabilities.

- ✓ The pmf: $p(x_i) = 1/n$

**BERNOULLİ RANDOM VARİABLE (1 trial)**

*Sonuçlarının bir başarı veya başarısızlık olarak sınıflandırılabilecek bir deneyin yapıldığını varsayalım.*

*(X represents the number of success that occur in n trials)*

- ➢ ***Success, p(1) = P(X=1) = β***
- ➢ ***Failure, p(0) = P(X=0) = $1 - \beta$***

***Where $\beta$ $(0 \leq \beta \leq 1)$ is the probability that the trial is a success.***

**BINOMIAL RANDOM VARIABLE (n trial) | *Binom ~ (n,p)***

Her birinin B olasılıklı bir başarı ile sonuçlanan ve 1-B olasılıklı bir başarısızlıkla sonuçlanan N bağımsız denemelerinin gerçekleştirileceğini varsayalım. Eğer X, N denemelerinde meydana gelen başarı sayısını temsil ediyorsa, X'in parametrelerle (N, p) binom rastgele bir değişken olduğu söylenir.

The pmf of a Binomial random variable with parameters (N, $\beta$) is given by

$$p(i)= \binom{N}{i} p^i(1-p)^{N-i}, \ i=0, 1, 2,.... N$$

Expected value: First moment; **E[X] = n. p**

Second moment; **E[X²]= n.p [(n-1).p +1]**

Variation: **Var[X]= n.p(1-p)**

**POISSON**

Poisson rastgele değişkeninin ağırlıklı olasılık değişkeni $\lambda$>0 parametresi ile birlikte

$$p(i) = P(X = i) = e^{-\lambda}\frac{\lambda^i}{i!}, \ i=0,1,....$$

**Expected Value: E[X] = $\lambda$**

**Variance: E[X²]= $\lambda(\lambda+1)$**

**Var[X] = E[X²]-E[X]² =$\lambda(\lambda+1)-\lambda^2=\lambda$**

Başka bir deyişle, her biri β olasılıkla başarı ile sonuçlanan N bağımsız denemeleri gerçekleştirilirse, gerçekleşen başarıların sayısı yaklaşık olarak bir Poisson rasgele değişkenidir.

**Örnekler:**

1) Bir kitabın sayfasındaki yanlış baskıların sayısı
2) Bir toplulukta 100 yaşına kadar hayatta kalan kişi sayısı
3) Her gün bir süpermarkette satılan köpek bisküvi paketlerinin sayısı

Poisson yaklaşımı, denemeler bağımlılıkları "zayıf" olduğu sürece bağımlı olmadığında da işe yarar.

**GEOMETRIC RANDOM VARIABLE**

Her birinin başarılı olma olasılığı independent olan bağımsız denemelerin bir başarı gerçekleşene kadar gerçekleştirildiğini varsayalım. X'in gerekli deneme sayısını göstermesine izin verin. X'in i'ye eşit olması için, ilk i-1 denemelerinin başarısızlıkla sonuçlanması ve i$^{th}$ denemenin başarılı olması gerekir.

**P(X=i) = (1- β)$^{i-1}$ . β , i=1,2,3,.....**

**E[X]= 1/ β**

**VAR[X] = $\frac{(1-\beta)}{\beta^2}$**

**NEGATIVE RANDOM VARIABLE**

Her birinin başarı olasılığı β, 0 <β <1 olan bağımsız denemelerin, toplam r başarı kazanılana kadar gerçekleştirildiğini varsayalım. X'in gerekli deneme sayısını eşitlemesine izin verirsek,

**P=(X=n) = $\binom{n-1}{r-1}\beta^r(1-\beta)^{n-r}, n = r, r+1, ....$**

**E[X] = r/$\beta$**

**E[X²]=$\frac{r}{\beta}\left(\frac{r+1}{\beta}-1\right)$**

**VAR[X]=$\frac{r(1-\beta)}{\beta^2}$**

**HYPERGEOMETRIC RANDOM VARIABLE**

Diyelim ki n boyutundaki örnek havuzundan rastgele seçilen bir numunesinin (değiştirilmeden), m topları beyaz ve N−m kadar siyah olan N topları içeren bir semaverden seçileceğini varsayalım. X'in seçilen beyaz top sayısını belirtmesine izin verirsek,

**P(X=i)= $\frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}}$ i=0,1,2,....**

**E[X]= $\frac{nm}{N}$**

**VAR[X] = $\frac{nm}{N}\left[\frac{(n-1)(m-1)}{N-1}+1-\frac{nm}{N}\right.$**

**RELATION BETWEEN PMF AND CDF**

**F(x)=$P(X \le x)$=$\int_{-\infty}^{\infty} f(t)dt$**

**EXPECTED VALUE OF CONTINUOUS RANDOM VARIABLE X**

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$\mathrm{Var}(X) = E\left[(X-\mu_X)^2\right] = E\left(X^2\right) - \mu_X^2$$

**Example:** Find the expected value and variance of $X$ whose pdf is given by

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_0^1 2x^2 dx = 2/3$$

$$E\left(X^2\right) = \int_0^1 2x^3 dx = 1/2$$

$$Var(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

**Proposition:** Assume $X$ is a continuous random variable. The expected value of $g(X)$ where $g(.)$ is a real-valued function is given by

$$E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Expectation is a linear operation.

$$E(aX+b) = aE(X)+b$$

Variance is not a linear operation.

$$Var(aX+b) = a^2 Var(X)$$

**RECITATION 4**

**Q1)** *An electronic scale in an automated filling operation stops the manufacturing line after three underweight packages are detected. Suppose that the probability of an underweight package is 0.001 and each fill is independent. (a) What is the mean number of fills before the line is stopped?*

***X: # of fills until system stops*** ***X~NegBin(0,001,3)***

***E[X]= r/p = 3/0,001=3000***

*(b) What is the standard deviation of the number of fills before the line is stopped?*

$$\sigma^2 = \frac{r(1-p)}{p^2} = \frac{3.(1-0,001)}{0,001^2} = 2997000$$

**Q2)** *A swimming pool repairman has* ***three check*** *valves in stock.* ***Ten percent*** *of the service calls require a check valve. What is the expected number and variance of the number of service calls she will make before using the last check valve?*

***Y~$NegBin(0,1,3)$***

***E[Y]=$\frac{3}{0,1} = 30$*** ***$Var[Y] = \frac{3.(1-0.1)}{(0.1)^2} = 270$***

**Q3)** *Assume that each of your calls to a popular radio station has a probability of 0.02 of connecting, that is, of not obtaining a busy signal. Assume that your calls are independent. (a) What is the probability that your first call that connects is your tenth call?*

***X~$Geo \ \ p = 0,02$*** ***P(X=10)= (1-0.02)$^9$. (0.02)= 0.0162***

*(b) What is the probability that it requires more than five calls for you to connect?*

***P(X>5)=$\binom{5}{0}(0,02)^0$ (1-0,02)$^5$=0.904***

*(c) What is the mean number of calls needed to connect?*

***E[X]=1/p=1/0,02=50***

**Q4)** *A continuous random variable T has the following probability density function.*

$$fx(u) = \begin{cases} 0, & u < 0 \\ 3\left(1-\frac{u}{k}\right), & 0 \le u \le k \\ 0, & u > k \end{cases}$$

*(a)*$\int_{-\infty}^{\infty} f_x(u)dx = 1 \rightarrow \int_0^k 3(1-\frac{u}{k})du = 1 \rightarrow k = \frac{2}{3}$

*(b)* ***E[T]=***$\int_{-\infty}^{\infty} uf_x(u)du \rightarrow \int_0^{2/3} u\left(1-\frac{3u}{2}\right)du = 2/9$

(c) $E[T^2]=\int_0^{\frac{2}{3}} u^2 f_x(u)du = \frac{2}{27}$

(d) $Var[T]= 2/27 - (2/9)^2= 2/81$

**Q5)** Let the random variable X denote a measurement from a manufactured product. Suppose the target value for the measurement is m. For example, X could denote a dimensional length, and the target might be 10 millimeters. The quality loss of the process producing the product is defined to be expected value of $k(X-m)^2 , where k is a constant that relates a deviation from target to a loss measured in dollars.

(a) Suppose that X is a continuous random variable with E(X)=m and V(X)=$\sigma^2$. What is quality loss of process?

$E[k(X-m)^2= k E[(X-m)^2] = k E[x^2-2xm+m^2] = k Var[x] = k \sigma^2$

**#QUIZ 2**

Answer questions 1-6 based on the following information:
A local newspaper company is printing papers using a printer with a cartridge that may break or run out of ink during operation. The cartridges can be replaced for 7.50 TL, but if they fail when the newsletter is being printed, the cost is estimated to be 25 TL because of the delay in publication. Let $X$ denote the number of weeks of use for a cartridge at the time of failure. The probability mass function ($p(.)$) of $X$ is:

| $X$ | $p(.)$ |
|---|---|
| 1 | 0.1 |
| 2 | 0.2 |
| 3 | 0.3 |
| 4 | 0.4 |

**1.)** Which of the following is closest to the value of $F(2.5)$?

$F(2.5)=P(X\leq 2.5) = p(1)+p(2) = \underline{0.3}$

**2.)** Which of the following is closest to the probability that the cartridge will be working during the first 2 weeks of use?

$P(X>2) = p(3)+p(4) = \underline{0.7}$

**3.)** Which of the following is closest to the expected value of the number of weeks of use before failure?

$E(X)= 0.1\times1+0.2\times2+0.3\times3+0.4\times4$
$= \underline{3}$

**4.)** Suppose the company decides to replace the cartridge immediately if it breaks or after every 3 weeks of use if it has not failed within this 3 weeks. Let $T$ denote cycle time, i.e., the elapsed time between installment of two consecutive cartridges. Which of the following is closest to the expected value of $T$?

$T= \min(X,3)$

$E(T)= E[\min(X,3)]= 0.1\times1+0.2\times2+(0.3+0.4)\times3$
$= \underline{2.6}$

**5.)** Suppose the company decides to replace the cartridge immediately if it breaks or after every 3 weeks of use if it has not failed within this 3 weeks. Let $T$ denote cycle time, i.e., the elapsed time between installment of two consecutive cartridges. Which of the following is closest to the variance of $T$?

$E(T^2)= E[\min(X,3)^2]=$
$= 0.1\times1^2+0.2\times2^2+0.7\times3^2= 7.2$

$Var(T)= E(T^2)-E(T)^2$
$= 7.2-2.6^2= \underline{0.44}$

**6.)** Suppose the company decides to replace the cartridge immediately if it breaks or after every 3 weeks of use if it has not failed within this 3 weeks. Let $Y$ denote the replacement cost in a cycle, where a cycle is defined as the elapsed time between two cartridges. Which of the following is closest to the expected value of $Y$?

$Y= 7.5\times I_{\{X\geq 3\}} + 25\times I_{\{X<3\}}$

$E(Y)= 7.5\times(0.3+0.4) + 25\times(0.1+0.2)$
$= 12.75$

where $I_{\{p\}} = \begin{cases} 1, & \text{if } p \text{ is true} \\ 0 & \text{o/w} \end{cases}$

**7.)** An insurance company studied the likelihood that different types of drivers in their database would be involved in at least one collision during any one-year period. All 20900 drivers involved in the database are categorized into four subgroups: teen, young adult, midlife, and senior. The results of the study shows that teen drivers have 15% chance to be involved in at least one collusion in any one-year period. The respective chances for young adult, midlife, and senior drivers are given as 8%, 4%, and 8% respectively. Out of 20900 drivers, 3836 are teens and 4463 are young adults. Which one is following is closest to the proportion of senior drivers so that the events "Any given driver has been involved in at least one collision in the past year" and "Any given driver is a young adult driver" are independent?

$C=\{$any driver has been involved in at least one collusion$\}$
$D=\{$any driver is a young adult$\}$
$P(D)= \frac{4463}{20900}$ $P(C|D)=0.08$
Let $x$ = proportion of senior drivers For independence, we need
$0.08=P(C|D) = P(C)= \frac{3836}{20900}\times0.15+ \frac{4463}{20900}\times0.08+(1-x-\frac{3836+4463}{20900})0.04 + x\times0.08$

$0.08=0.0446+(1-x-0.397)\times0.04 +0.08x$
$x= \underline{0.281722}$

**8.)** In how many of the following cases a discrete random variable is more appropriate to use as compared to a continuous one?

- The number of heads in 100 tosses of a coin. (Discrete)
- The exact length of a rod randomly chosen from a day's production. (Cts)
- The final exam score of a randomly chosen student from Quiz #1 of this semester's MATH 215 class. (Discrete)
- The age that a randomly chosen OzU student will be on his or her next birthday. (Discrete)

**Discrete: Sayılan (number)**

**Continuous: Ölçülen (lenght, weight)**

Answer questions 9-13 based on the following information:
A traffic light at a certain intersection is green 50% of the time, yellow 10% of the time and red 40% of the time. A car approaches this intersection once each day. (Reminder: You can pass cautiously if the light is yellow.)

**9.)** Which one is following is closest to the probability that the car will have to stop a total of 5 times in this traffic light over the next 20 days?

$X\sim Bin(20, 0.4)$
$P(X=5)= \binom{20}{5} 0.4^5\, 0.6^{15} = \underline{0.074647}$

**10.)** Which one of the following is closest to the standard deviation for number of times the car will have to stop over the next 20 days?

$Var(X)= N.\beta.(1-\beta)= 20\times0.4\times0.6$
$= 4.8$

$SD(X)= \sqrt{4.8} = \underline{2.19}$

**11.)** Which one of the following is closest to the probability that more than 3 days will pass up to and including the first time the car encounters a red light?

$Y\sim Geo(0.4)$
$P(Y>3)= 1-P(Y\leq 3)= 1-P(Y=1)-P(Y=2)-P(Y=3)$
$= 1-0.4-0.4\times0.6-0.4\times0.6^2$
$= \underline{0.216}$

**12.)** Which one of the following is closest to probability that it will take 7 days on which a red light is encountered for the third time?

$Z \sim \text{Neg Bin}(3, 0.4)$

$P(Z=7) = \binom{6}{2} 0.4^3 \, 0.6^4 = \underline{\underline{0.1244}}$

**13.)** Which one of the following is closest to the variance for the total number of days such that the car will pass through the traffic stop without stopping for the seventh time?

$W \sim \text{Neg Bin}(7, 0.6)$

$Var(W) = \frac{r \cdot (1-B)}{B^2} = \frac{7 \times 0.4}{0.6^2} = \underline{\underline{7.77}}$